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Plural and feminine marking in Lunigianese varieties

A(n apparent) case of phonologically-determined affix reordering

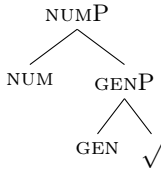
1 Introduction

According to the Mirror Principle (Baker, 1985, 375), “morphological derivations must directly reflect syntactic derivations (and vice versa)”. This suggests that the linear ordering of morphemes is determined by syntax, and that the two derivations are fundamentally the same, namely that “morphology is syntax all the way down” (Halle & Marantz, 1993; Halle, 1997; Brody, 2000; Manzini & Savoia, 2007; Starke, 2009; Collins & Kayne, 2020; Caha et al., 2025). One way of implementing this idea is adopting Cinque (2005)’s analysis of U20 and stating that the linear ordering of morphological markers reflects the hierarchical structure of the universal functional spine (Cinque, 1999; Rizzi, 2004), or is derived from the former by applying a constrained set of movement operations to base-generated structures (Cinque, 2005; Taraldsen, 2010; Caha, 2012; Muriungi, 2014; Myler, 2017). Let us see how this applies to nominal expressions displayed by Romance languages.

It is relatively uncontroversially assumed that nominal expressions are base-generated by merging a root first with a GENDER head (henceforth GEN)¹, and then with a NUMBER head (henceforth NUM; Ritter, 1991; Carstens, 2000; Koopman, 2003; Carminati, 2005; Fuchs et al., 2016; Baggio, 2025 a.o.), yielding the representation in (1).

- (1) NUMP syntactic structure

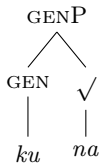
¹ I adopt a broad definition of GEN, which blurs the distinction, if there is any, between GEN and CLASS, and maintains that GEN, as CLASS, is a feature that determines concord properties and groups nouns into concord classes (Hockett, 1958; Carstens, 1993; Kihm, 2005; Arsenijević, 2018; Fuchs & van der Wal, 2022).



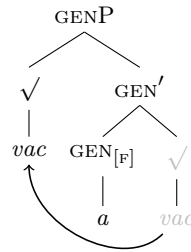
Let us now assume that, once such structure has been generated, $\checkmark$ can optionally move upwards pied-piping its complement, and that, when it does apply, movement is total (Cinque, 2005). In such a scenario, if movement does not apply, the linearization of the hierarchical structure faithfully reflects the hierarchical ordering of syntactic heads, namely NUM-GEN- $\checkmark$, whereas if movement does apply, $\checkmark$ and its pied-piped material move all the way up to Spec,NUM, resulting in a linear ordering that is the mirror image of the syntactic structure, namely $\checkmark$ -GEN-NUM.

These two linearization patterns are displayed by Jaad (Pozdniakov, 2013) and Spanish. In (2) and (3) we can see the syntactic structure of the forms for ‘cow’ in Jaad and in Spanish, respectively. In Jaad, the linear ordering of the the markers – $ku_{\text{GEN}}-na_{\checkmark}$ – faithfully reflects the base-generated syntactic structure, i.e. $[\text{GEN}[\checkmark]]$, whereas in Spanish, the linear ordering of the markers – $vac_{\checkmark}-a_{\text{F}}$ – is the mirror image of the order of the base-generated syntactic structure. As shown in (3), this can be derived by moving the root vac in Spec,GENP.²

(2) Jaad $ku-na$ ‘cow’



(3) Spanish $vac-a$ ‘cow’



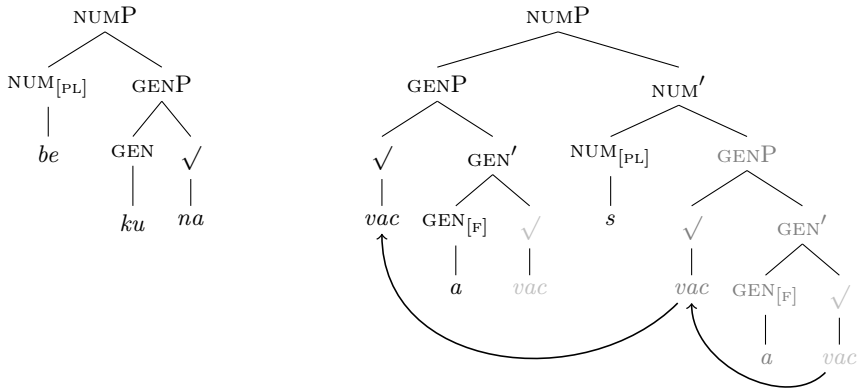
The same asymmetry can be observed if we add the PL marker to the structures in (2) and (3). As shown in (4), in Jaad, the base-generated ordering of heads –

² In order to emphasise the parallelism between the Jaad and the Spanish examples, I represent FEMININE (henceforth F) as a value of GEN; however, as discussed later, I maintain that F is actually a privative feature heading its own projection (see Caha, 2022; Baggio, 2025 a.o.); the same holds for PLURAL (henceforth PL) (Caha, 2022). In these and other syntactic derivations, unpronounced material is greyed out.

[NUM[GEN[$\sqrt{\quad}$]]] – is faithfully reflected by the ordering of the markers: *be_{PL}-ku_{GEN}-na _{$\sqrt{\quad}$}* . In Spanish, the linear ordering of the markers – *vac _{$\sqrt{\quad}$} -a_F-s_{PL}* – is the mirror image of Jaad’s. This can be obtained by cyclically moving $\sqrt{\quad}$ in Spec,NUMP pied-piping GENP (5).

(4) Jaad *be-ku-na* ‘cows’

(5) Spanish *vac-a-s* ‘cows’



Although the Mirror Principle is considered a robust generalisation, exceptions are not unheard of. Some of them have been explained by appealing to (language-/construction-/head-specific) syntactic non-pied-piping movement operations, which strand some of the affixes that are cyclically moved through by the lexical root (Taraldsen, 2010; Muriungi, 2014; Myler, 2017). Alternatively, Mirror Principle exceptions have been argued to be due to arbitrary morphological templates (Bloomfield, 1962; Zwicky, 1985; Anderson, 1986; Simpson & Withgott, 1986; Speas, 1990; Stump, 1991; Inkelas, 1993; Hyman & Inkelas, 1999; Good, 2003), postsyntactic operations such as Local Dislocation (Embick & Noyer, 2001; Arregi & Nevins, 2012), weighted bigram constraints encoding morphotactic restrictions (Ryan, 2010), specific rankings of morphophonological alignment and phonological constraints (Kim, 2010; Zukoff, 2023), or phonological representations (Paster, 2009; Kim, 2015; McCann, 2025). This seems to suggest that syntax is possibly not the only module responsible for the linear ordering of morphosyntactic constituents (see Rice, 2011; Manova, 2015 for an overview of semantic, syntactic, morphological, and phonological factors that possibly interact in determining affix ordering).

In what follows, I provide further support for the role played by phonological representations in what looks like a case of affix ordering, focusing on the Mirror Principle-violating ordering of the F and PL markers in a set of Romance variety spoken in Lunigiana (Italy), where the PL marker occurs between the lexical root and the F marker, as in Colonnatese *don-j_{PL}-a_F* ‘women’. More precisely, I show that such linear ordering does not require any ad hoc syntactic derivation, arbitrary

morphological template, postsyntactic operation, bigram constraint, modularity-violating interaction of alignment and phonological template, nor the postulation of several NUM heads in the nominal functional sequence (Manzini & Savoia, 2017). In fact, I argue that if we build on independent phonological properties of the relevant varieties, and refine the phonological representation of the F and PL markers, an alternative analysis can be devised in which the violation of the Mirror Principle is only apparent, and the relevant syntactic structure is exactly as those of other Romance varieties, where nominal expressions have the structure in (1).

The soundness of such syntactic structure, and, indirectly, of the proposed analysis, is supported by another property of the relevant Lunigiana varieties, i.e. partial concord (henceforth PC; also known as ‘lazy’ or ‘asymmetrical’ concord; Beninca & Haiman 1992; Rasom 2008; Pomino 2012; Bonet 2018; Cyrino & Espinal 2020; Manzini et al. 2020; Pescarini 2021). More precisely, the hypothesis that Lunigianese nominal expressions are derived from a structure such as that in (1), where NUM is merged higher than GEN, is supported by the fact that, when PC applies, the missing marker is PL. Crucially, this militates against analyses in which the observed $\sqrt{\text{-PL-F}}$ linear ordering is due to NUM merging lower than GEN (Manzini & Savoia, 2017), which, if correct, would preempt the proposed phonology-based analysis. Lunigianese PC is shown below with data from the varieties spoken in Colonnata and Bagnone. Both these varieties have *lup-j_{PL-F}-a_F* bare NPs – (6), (8) – but if an article enters the derivation, the PL marker surfaces either on the article, as in Colonnatese (7), or on the noun, as in Bagnonese (9).

- | | |
|---|---|
| (6) <i>lup-j-a</i>
wolf-PL-F
‘wolves’ | (8) <i>lup-j-a</i>
wolf-PL-F
‘wolves’ |
| (7) <i>l-j-a lup-a</i>
the-PL-F wolf-F
‘the wolves’ | (9) <i>l-a lup-j-a</i>
the-F wolf-PL-F
‘the wolves’ |

According to the Concord Hierarchy Generalization (Bayırılı, 2017; Caha, 2022), if, in a specific language, a functional or lexical element shows concord for CASE (henceforth K), it shows concord for NUM and GEN too, and if it shows concord for NUM, it shows concord for GEN too. The other way around is not true. This provides support for i) a syntactic structure like that in (1) (for [K [NUM]] see also Greenberg, 1963; Klaudová, 2020), and ii) the hypothesis that “different types of [partial] concord [correspond to] different structures that are trimmed top down”, namely that “categories can be missing from the top of that hierarchy, but not

in the middle” (Caha, 2022).³ Crucially, this confirms that the missing marker in Lunigianese PC structures, despite the fact that it looks linearly closer to the root than GEN, is actually hierarchically higher, namely, it is the one sitting on the top of the hierarchy, which gets trimmed away under PC. This, in turn, corroborates the analysis proposed in this paper, where the Mirror Principle violation is a phonology-induced hallucination, as no syntactic nor morphological operation applies and changes the hierarchical ordering of the GEN and NUM affixes.

On a more general theoretical level, I maintain that the patterns displayed by these Lunigianese varieties provide an ideal empirical grounding on which to compare and decide upon i) interface theories that allow for a high degree of interaction between (morpho)syntax and phonology (Jackendoff, 2007; van Oostendorp, 2007; Boersma, 2011; Trommer, 2015; Zukoff, 2023), as opposed to theories that do not admit any interaction and consider phonology a purely interpretative module (Chomsky, 2008; Starke, 2009; Scheer, 2012), and ii) theories proposing different formal identities of the NUM head and (the domain of) concord operation (Piccolo, 1991; Cardinaletti & Giusti, 2015; Manzini & Savoia, 2017; Cyrino & Espinal, 2020; Manzini et al., 2020; Pescarini, 2021; Wiltschko, 2021). In this respect, I contend that these data favour an interface theory where there is no direct interaction between syntax and phonology, and a syntactic theory where NUM has the formal status of a head.

The details of the phonological analysis, presented in section 3.2, are preceded by a brief discussion of an alternative syntax-based analysis (Taraldsen, 2010) and of one resorting to postsyntactic operations (section 3.1). This is followed by a short section on PC (section 3.3) and by a concluding section (section 4). Let us start, though, by providing more details of the GEN and NUM marking in the relevant Lunigianese varieties.

2 Data

Lunigianese varieties are a group of non-standard Gallo-Romance languages spoken across the boundaries of Liguria and Tuscany (Italy), in a geolinguistic domain characterized by an extremely high degree of microvariation. As shown by the existing literature, this microvariation concerns both the phonetic, phonological

³ Note that there seem to be counterexamples to the Concord Hierarchy Generalization, e.g. in Georgian, Khwarshi, Zulu and Dime. I refer the reader to Caha (2022) for an analysis of such patterns that show that, in fact, a nanosyntactic analysis that builds on root size is compatible with the Concord Hierarchy Generalization.

and morphosyntactic domains (Restori, 1892; Bottiglioni, 1911; Giannarelli, 1913; Maffei Bellucci, 1977; Savoia, 1983; Loporcaro, 1994; Carpitelli, 1995). Most of the studies focusing on the linguistic microvariation of this area are mainly concerned with the description of lexis, pronunciations and/or syntactic constructions. Besides a few sections in Manzini & Savoia (2005), no serious attention has been paid to the grammatical systems underlying the observed surface variation.

Despite the high degree of microvariation, these varieties share several common features. Among the most relevant for us, there is the fact that these varieties have a rather impoverished inflectional gender system. Nouns belonging to the Latin I and III F declension merged into one F class, characterised, in the singular, by the *-a* inflectional marker (e.g. CARNĚ(M) ‘meat’ > Colonnatese [ˈkarna], cfr. It. [ˈkarne]). Similarly, nouns belonging to the Latin II and III M declension merged into one M class, which, in the singular, shows no marker (e.g. PISCĚ(M) ‘fish’ > Colonnatese [ˈpeʃ], cfr. It. [ˈpeʃːe]). The PL of this class is lexicalised by the *-i* marker (Colonnatese [ˈpeʃi] ‘fish.PL’), while F.PL nouns display both the marker of PL and that of F, i.e. *-i-a*, surfacing as [-ja].⁴ The Latin N.PL marker *-a* disappeared, and the members of that class merged with other M nouns, and now show the *-i* PL marker (BRĀCHĬA ‘arms’ > Colonnatese [ˈbratːʃi], cfr. It. [ˈbratːtʃa]). In Lunigianese varieties, nouns belong to either one or the other of these two classes.⁵ The distribution of the relevant markers is summarised in table 1 with the forms for ‘wolf’.

Tab. 1: Inflectional classes

	SG	PL
M	lup	lup-i
F	lup-a	lup-j-a

⁴ Note that *-i* is a shorthand for the actual underlying representation, which, as discussed later, correspond to the element [I] both in F and M forms. Note also that other Lunigianese varieties either lack a dedicated marker for F.PL (e.g. Pontremolese *al gat* ‘the cats’), or lexicalise it as in Standard Italian, i.e. as *-e* (e.g. Carrarese *le gate* ‘the cats’).

⁵ Another inflectional class could possibly be identified which derives from Latin IV declension. This class contains a small set of F nouns that, in Latin, end in *-us*. This marker shares the same evolution of M *-us*, namely *-us* > *-o* > $\emptyset$. Despite the absence of the corresponding marker on the noun, F is retained. This is shown e.g. by Colonnatese *la man* ‘the hand’, where the F marker is suffixed to the article (cf. also *lja man* ‘the hands’). I assume that the behaviour of this class can be adequately captured by a nanosyntactic analysis, in which the root *man* reaches the PLP size.

As already mentioned (see e.g. (7) and (9)), in Lunigiana varieties, nominal expressions are characterised by PC. In PC structures, the F marker occurs in all the functional and lexical elements of the nominal expression, whereas the PL marker occurs only in a subset of them (Loporcaro, 1994; Manzini & Savoia, 2005; Cavarani, 2018). For instance, if an adjective or a pronoun enters the nominal expression, the PL marker can surface either only on the article, as in the Colonnatese examples in (10), (11), and (12), or, as shown by the examples from Bagnonese, on the constituent occurring in second position, no matter whether the latter is an adjective (13), a pronoun (14), or a noun (15).

- | | |
|---|---|
| <p>(10) <i>l-j-a bɛl-a lup-a</i>
 the-PL-F beautiful-F wolf-F
 ‘the beautiful wolves’</p> | <p>(13) <i>l-a bɛl-j-a lup-a</i>
 the-F beautiful-PL-F wolf-F
 ‘the beautiful wolves’</p> |
| <p>(11) <i>l-j-a nostr-a lup-a</i>
 the-PL-F our-F wolf-F
 ‘our wolves’</p> | <p>(14) <i>l-a nostr-j-a lup-a</i>
 the-F our-PL-F wolf-F
 ‘our wolves’</p> |
| <p>(12) <i>l-j-a lup-a ner-a</i>
 the-PL-F wolf-F black-F
 ‘the black wolves’</p> | <p>(15) <i>l-a lup-j-a ner-a</i>
 the-F wolf-PL-F black-F
 ‘the black wolves’</p> |

Maffei Bellucci (1977) and Luciani (2003) claim that the PL marker tends to be realized only once, “almost always in conjunction with the first nominal form”. When more instances of PL occur (e.g. Colonnatese *tant-j-a donn-j-a* ‘many-F-PL woman-F-PL’), the repetition is “perceived as redundant”, rather than as “the violation of a norm”. As shown in Cavarani (2018), the PL marker is actually repeated in many more contexts than previously thought, and show a high degree of inter- and intra-linguistic microvariation. For instance, whereas young Colonnatese speakers tend to realise the PL marker only once, older speakers tend to show complete concord. This is in line with the diachronic trajectory proposed by Rasom (2008) for Gardenese, Friulian and Central Ladin, and by Sauzet (2012) for Occitan. Something similar (albeit in the opposite direction) is also reported for the varieties of Bagnone and Treschietto by Loporcaro (1994), who observes that the variation he found almost 20 years after Maffei Bellucci (1977)’s fieldwork could be interpreted as an internal evolution of these varieties, which apparently tends to extend the realization of the PL marker. The same is maintained by Pescarini (2021) for a set of Lombard Alpine varieties.

Note that the few forms given above are just a selection of the marking options observable in Lunigiana varieties. As soon as more varieties and nominal expressions containing different constituents are taken into consideration, the range of marking options drastically increases. A wide array of these options is shown in Cavarani

(2018), which presents data collected by means of a literature survey and fieldwork session in all the location showing PC (Bedizzano, Bergiola, Colonnata, Ameglia, Arcola, Santo Stefano di Magra, Bolano, Villafranca in Lunigiana, Treschietto, Iera, Groppo, Nezzana, Pieve, Filetto, Filattiera, Caprio, Gigliana, Lusignana, Rocca Sigillina, Ponticello, Tresana, Mulazzo). Interestingly, the fieldwork shows that, in some varieties, the PL marker can fail to surface altogether. Despite being absent from the phonetic signal, though, the PL head is present at the level of syntactic representation, which explains e.g. why a PL subject that shows no PL marker triggers PL agreement on the verb. According to Wiltschko (2021), the possibility of triggering verb agreement suggests that PL has the status of a syntactic head. I thus maintain that, despite its optional realization, PL is a syntactic head in Lunigianese varieties too. This optional realization is shown below, where the PL marker surfacing on the demonstrative in (16) can be left unpronounced (17), but still triggers verb agreement.

- (16) st-j-a pegr-a l εŋ l-a nɔstr-a
 this-PL-F sheep-F SCL be.3PL the-F our-F
 ‘these sheep are ours’
- (17) kl-a pegr-a l εŋ l-a vɔstr-a
 that-F sheep-F SCL be.3PL the-F your-F
 ‘those sheep are yours’

A first step towards the complete characterisation of the PC microvariation is Cavirani (2018) (see also Maffei Bellucci, 1977; Loporcaro, 1994; Cuneo, 2001; Manzini & Savoia, 2005), but as the whole set of data still need to be analysed, I am not going to propose an analysis of PC (see Pomino, 2012; Cyrino & Espinal, 2020 for an analyses based on a subset of the data; see also section 3.3).

Another relevant property shared by all Lunigiana varieties is unstressed vowel reduction (Cavirani, 2015). In most varieties, Latin unstressed vowels reduced to schwa or were completely deleted, depending on the position in the string and the quality of the vowel, with protonic and corner vowels resisting deletion more than postonic and mid vowels. Besides in a very small set of varieties, the only postonic vowels that are retained are *i* and *a* occurring in final position, namely those that corresponding to the PL and F markers. The mid vowels that in a diachronically preceding stage corresponded to M and F.PL, i.e. *o* and *e*, respectively, were elided. Note that unstressed vowel reduction is still a productive process, as shown by alternations such as Carrarese [u'zɛd] ‘bird’ ~ [uz(ə)'diŋ] ‘little bird’, where the root vowel /ɛ/ is reduced to schwa or elided due to the movement of stress triggered by the diminutive suffix /'iŋ/. Note also that these varieties generally lack sequences

of unstressed vowel-glides – such as /aj/ – in final position (Cuneo, 2001), whereas glide-vowel sequences – such as /ja/ – are widely attested.

As I show in the next section, by combining what we know about the phonology of these varieties and about PC, and resorting to a specific formal modelling of phonology, we can provide a straightforward account of the unexpected linear ordering of the F and PL markers.

3 Formal analysis

3.1 Morphosyntax-based analyses

One way to analyse the unexpected linear ordering of the F and PL markers in Lunigianese varieties is arguing that there are two structural position for NUM (Lecarme, 2002; Landau, 2016; Lahrouchi & Lampitelli, 2014; Kramer, 2016). Manzini & Savoia (2017) claim that this is what Friulian data like those in (18)–(21) suggest.

- (18) l-i bjel-i fēmin-i-s
 the-PL nice-PL woman-PL-PL
 ‘the nice women’
- (19) l-i fēmin-i-s vɛtʃ-i-s
 the-PL woman-PL-PL old-PL-PL
 ‘the old women’
- (20) l-a altr-a fēmin-a
 the-F other-F woman-F
 ‘the other woman’
- (21) ke-i bje-i ɔŋ-s vɛtʃ-u-s
 that-PL fine-PL man-PL old-M-PL
 ‘those fine old men’

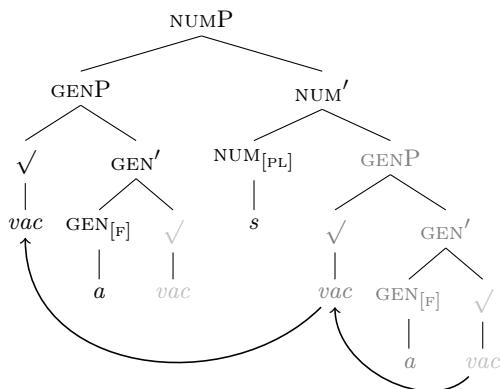
In Friulian, the fact that i) PL determiners and pronominal adjectives show a *-i* marker, ii) PL nouns show an additional marker, i.e. *-s*, and iii) SG forms do not show such markers, seem to suggest that there indeed are two syntactic positions for PL.⁶ This does not seem to be the case in Lunigianese varieties. To the best of my knowledge, Romance varieties have been claimed to have a high (sigmatic) PL,

⁶ Though see Bonet (2018), who suggests that *-i* is an allomorph of F that surfaces only in the context of PL, as possibly indicated by the absence of *-i* from the PL noun ɔŋ-s in (21).

or both a high and a low PL (this is possibly the case of *-a* forms such as Italian *uov-a* ‘eggs’, but see Loporcaro & Paciaroni, 2011), but never a low PL alone. If this is true, and if the *-j* marker occurring between the root and the F marker in Lunigianese varieties were the lexicalisation of a low PL, we would be led to conclude that either these varieties do not have a high PL, or that the *-a* marker occurring on the right of *-j* is itself the marker of a high PL. If that were the case, we would expect to find occurrences of high PL alone, namely PL forms with the *-a* marker alone, which is not what we find (as mentioned above, there is not forms such as It. *uov-o~uov-a* ‘egg’~‘eggs’), and we would need to explain why M forms have a low PL only. Furthermore, low PL (e.g. the Italian *-a* type, according to Manzini & Savoia, 2017) is only partially productive and semantically idiosyncratic (Acquaviva, 2008), which is not the case of *-j/-i* PL in Lunigianese varieties. I conclude that the hypothesis that Lunigianese varieties’ *-j* marker lexicalises a low PL head is implausible.

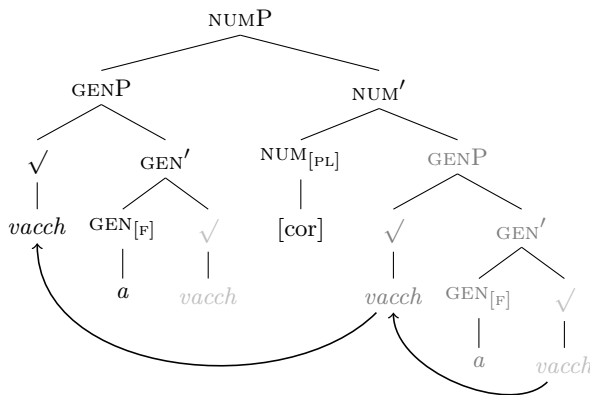
Another way to analyse the Lunigianese linear ordering of the F and PL markers is arguing that, after a relatively standard roll-up derivation of NUMP, a postsyntactic operation applies at PF that lowers and right-adjoins the GEN marker to the NUM marker. The fact that, in Romance, NUMP is derived via roll-up movement is clearly shown by western Romance varieties: one of the distinguishing properties of these varieties is that they show a sigmatic PL marking, which linearly follows the GEN exponent. This is what we see e.g. in Spanish and Portuguese *cas-a~cas-a-s*, Catalan *cas-ə~cas-ə-s*, Romansh *chas-a~chas-a-s*, Sardinian *dom-o~dom-o-s*, Occitan *ostal~ostal-s*, (orthographic) French *maison~maison-s* ‘house’~‘houses’ (Pomino, 2017). A syntactic derivation of western Romance NUMP is show in (22), repeated from (5), with the Spanish form for ‘cows’.

(22) Spanish *vac-a-s* ‘cows’



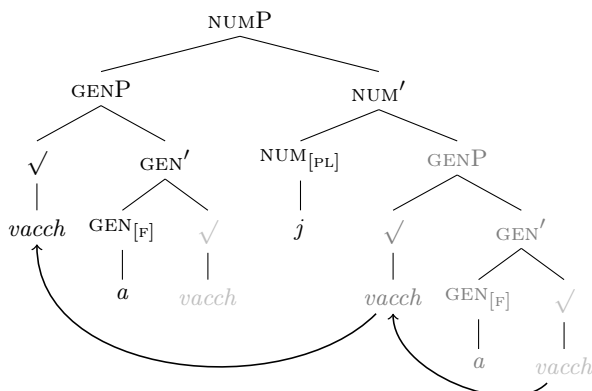
As shown, once GEN is merged to $\checkmark$, the latter moves to Spec,GENP, which in turn moves to Spec,NUMP once GEN is merged. This results in *vac-a-s*. Interestingly, the same kind of movement operations have been argued to derive eastern Romance NUMP too, despite the latter do not seem to show a marker dedicated to PL, but rather a change in the quality of the vocalic marker following the root, as in It. *cas-a*~*cas-e*, Sicilian *cas-a*~*cas-i* and Romanian *cas-ă*~*cas-e* ‘house’~‘houses’ (Pomino, 2017). In fact, on an abstract level, all Romance languages can be argued to use morphological augmentation as a NUM marking strategy (d’Hulst, 2006). This hypothesis can be maintained if we assume that whereas in western Romance varieties PL is lexicalised by a sigmatic marker that surfaces independently, in eastern Romance PL is lexicalised by a floating phonological feature that associates to the segment marking GEN, modifying the quality of the latter (Passino, 2009; Lampitelli, 2010, 2014; Lampitelli & Ulfsbjorninn, 2023). This is shown in (23) with the Italian form for ‘cows’, where GEN is lexicalised by a full vocalic segment – *a* – whereas PL is lexicalised by a floating [coronal] feature.

(23) Italian *vacch-e* ‘cows’



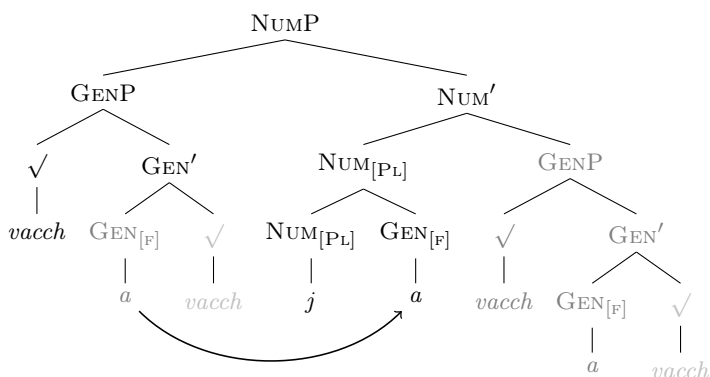
Once this structure is sent to phonology, the string /vakɪ-a-[cor]/ surfaces as ['vakɛ] due to the fronting of /a/ induced by [cor]. At first sight, an approach like this would not seem to work to explain the $\checkmark$ -*j*-*a* linear ordering of Colonnatese varieties, as the NUM marker looks like a full segment that infixes between the root and the GEN marker, rather than a floating feature that associates to the GEN marker. Thus, one might want to propose that, like virtually all the other Romance varieties, Colonnatese syntax generates the structure in (24).

(24) Colonnatese *vacch-j-a* ‘cows’ - syntactic structure



Once this structure is generated, a postsyntactic operation such as Local Dislocation (Embick & Noyer, 2001) would apply which lowers and right-adjoins the GEN head to the NUM head deriving Colonnatese [vak:ja] ‘cows’ (25).

(25) Colonnatese *vacch-j-a* ‘cows’ - Local Dislocation

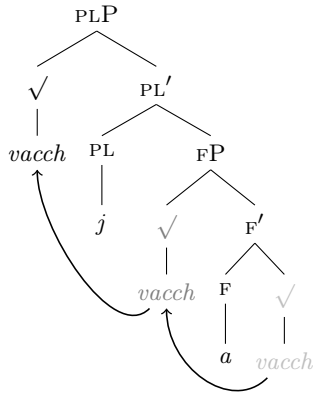


Despite Local Dislocation can generate the observed linear ordering of Colonnatese F and PL markers, no other motivation than producing the latter can be adduced to support the hypothesis that Local Dislocation applies. Furthermore, Lunigiana varieties would be the only Romance variety in need for such a postsyntactic operation. I take it to suggest that there could be an alternative, more plausible analysis, which does away with Local Dislocation and with the independent morphological component it assumes.

An alternative analysis is the one developed in Taraldsen (2010), which provides an account of Colonnatese F and PL marker ordering within the nanosyntax framework (Starke, 2009; Caha et al., 2025). In this analysis, the linear ordering of the F and PL markers is derived fully syntactically by assuming that the lexical

root moves cyclically to Spec,FP and Spec,PLP without pied-piping. This is shown in (26), where, in line with nanosyntactic assumptions, each syntactic feature heads its own projection, and features are privative (Taraldsen, 2010, 115).

(26) Colonnatese *vacch-j-a* ‘cows’ - nanosyntax



The derivation in (26) is interpreted by phonology as [vak:ja], capturing the observed linear ordering of Colonnatese F and PL markers. It does that without stipulating postsyntactic operations such as Local Dislocation, thus without stipulating an additional grammar module, i.e. morphology. In this sense, Taraldsen (2010)’s analysis is more parsimonious and more restricted than a Distributed Morphology-like analysis. However, I maintain that this comes with a price, which is assuming that the relevant Lunigianese varieties stand out with respect to all the other Romance varieties as the only ones that derive nouns without roll-up upwards movement.⁷ Note, also, that there is no evidence for such type of movement to apply in other corners (e.g. in the verbal system) of the grammar of Lunigianese varieties. For this reason, I propose an alternative account, which does not require postsyntactic operations, nor a non-standard syntactic derivation.

3.2 A phonology-based analysis

The phonological analysis I am proposing resorts to the formal tools provided by Strict CV (Lowenstamm, 1996; Scheer, 2004) and Element Theory (Bakkey,

⁷ In fact, Taraldsen (2010) argues that Colonnatese has one nominal class which is characterised by a roll-up movement derivation, and one where the lexical root moves without pied-piping. I believe that this adds further unnecessary complexity to the analysis.

2011). If we adopt such tools, we can formalise the $\sqrt{-j_{PL}-a_F}$ linear ordering of Lunigianese varieties as a case of segmental re-ordering (Paster, 2009). Crucially, this re-ordering is within the exclusive purview of phonology, as it follows from the phonological representation of the relevant markers (see McCann, 2025 for a similar proposal, albeit in a radically different theoretical framework).

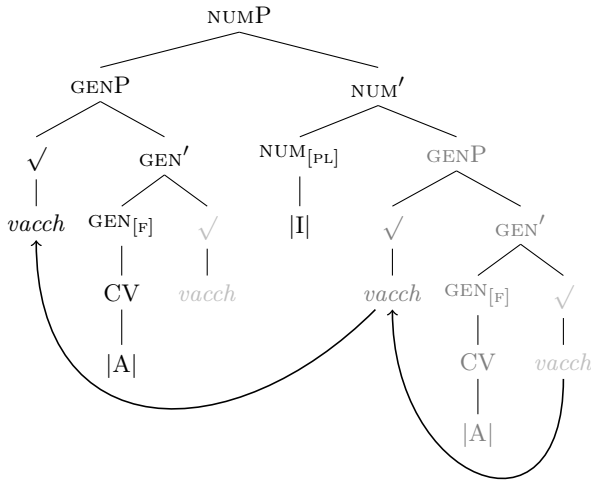
Let us start by recalling two properties characterising the phonological system of the varieties displaying the relevant linear ordering. The first is the fact that, in these varieties, unstressed mid vowels are reduced, namely, they surface as schwa, or delete. Thus, unless protected by stress, underlying mid vowels do not surface as such. The second fact is that unstressed falling diphthongs are also virtually absent, whereas rising diphthongs/consonant clusters are very frequent. In Element Theory terms, the first property can be formalised as the inability of Lunigianese unstressed vowels to license elemental expressions (Cavirani, 2015). In other words, unstressed nuclei prefer to contain as little elements as possible. This is shown in (27) with the Colonnatese forms for ‘bird’ and ‘little bird’ (I provide the element structure of the relevant segments only).

(27) Colonnatese [u'ʒɛd]~[uʒ'diŋ] ‘bird’~‘small bird’

a. C V 'C V C V → b. C V C V 'C V C V
 | | | | | | | | | |
 |U| ʒ |AI| d |U| ʒ d |I| ŋ

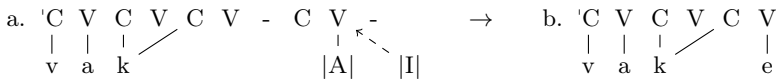
As per Strict CV, the phonological representation of the forms in (27) displays an initial empty C slot, and a final empty V slot. The first vocalic segment – [u] – is represented as a V slot associated to a |U| element, and the second – [ɛ] – as a V slot associated to the |AI| complex expression. As shown in (27)b, when a marker that shifts the stress is added, the mid vowel drops. This is arguably due to the inability of unstressed nuclei to license phonological information (van Oostendorp, 1995; Harris, 1997, 2005; Cavirani, 2015, sub). As discussed above, this process affects to the highest degree postonic nuclei and mid vowels. This suggests that whenever the grammar of these varieties requires two vocabulary items containing the elements |A| and |I| to combine, the two elements cannot be associated to one and the same unstressed V slot. As argued by Passino (2009); Lampitelli (2010, 2014); Lampitelli & Ulfsbjorninn (2023), such a combination obtains e.g. in Italian when the $[[[\sqrt{\quad}] F] PL]$ syntactic structure is lexicalised. This structure is shown in (28) with the Italian form for ‘cows’, where the F and the PL markers are represented, respectively, as a full segment, i.e. a CV unit with the element |A| associated to the V slot, and a floating |I| element, namely an element that is not associated to any CV slot.

(28) Italian *vacch-e* ‘cows’



As shown, the syntactic derivation is identical to that of a Spanish NUMP given in (5), and arguably to that of all Romance NUMP, and features the lexical root cyclically moving to Spec,GENP, and then to Spec,NUMP, pied-piping GENP. Differently from Spanish, though, the PL marker is represented as a floating |I| element. This means that, in order surface, it needs to associate to a CV slot. Crucially, the F marker provides the floating |I| a slot to associate to, namely the V slot containing floating |A|. Thus, once the syntactic structure is sent to the phonological module, the floating |I| associates to the first available host on its left, namely to the V slot of the F marker, deriving |AI|, i.e. [e]:⁸

(29) Italian /'vak:-a-i/ → ['vakɐ] ‘cows’

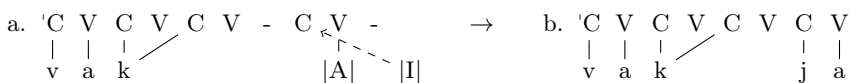


As Italian unstressed V nodes can license complex elemental expressions, associating two elements in the same V node creates no problems. As we have seen,

⁸ Following Gussmann & Kaye (1993), I assume that empty VC sequences get deleted at the end of the derivation. In (29), this sequence is the one occurring at the ✓-F boundary. As shown below in (30), this deletion operation does not apply in Colonnatese because the C slot of the F marker hosts |I|.

though, this is not the case in Lunigianese varieties, as they do not allow complex element expression in unstressed V nodes. Thus, assuming that Lunigianese PL marker is identical to Italian PL marker, its association to the first available host on its left – i.e. the F marker's V node – is blocked. |I| thus tries to associate to the next available host, i.e. the empty C node of the F marker. Assuming the standard autosegmental hypothesis that consonants and vowels sit on different tiers, and that |I| is not underlyingly associated to any CV slot, namely, it is neither a consonant nor a vowel, associating to the C slot of the F marker does not involve any line crossing violation. Hence, as there is nothing preventing |I| associating to the empty C node of the F marker, |I| surfaces as [j] to the left of |A| [a], as shown in (30).

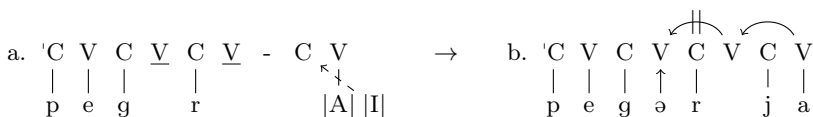
(30) Colonnatese /'vak:-a-i/ → ['vak:ja] 'cows'



Note that, whatever the reason is why falling diphthongs are absent from these varieties⁹, the proposed representations of the F and PL markers derives the fact that the |I| has no chance of surfacing to the right of |A| – generating *ai/aj* – for the F marker's V node hosting |A| is followed by no available C/V host.

The hypothesis that |I| associates to a C slot – and as such pronounced as [j] – is supported by forms such as Bagnonese ['pegərja] 'sheep.PL', which features an epenthetic [ə] between the radical [g] and [r] (cf. ['pegra] 'sheep.SG'). As shown in (31), once the F marker is concatenated and the string is parsed by phonology, the resulting structure displays two empty V slots in a row (underlined).

(31) Bagnonese /'pegr-a-i/ → ['pegərja] 'sheep.PL'



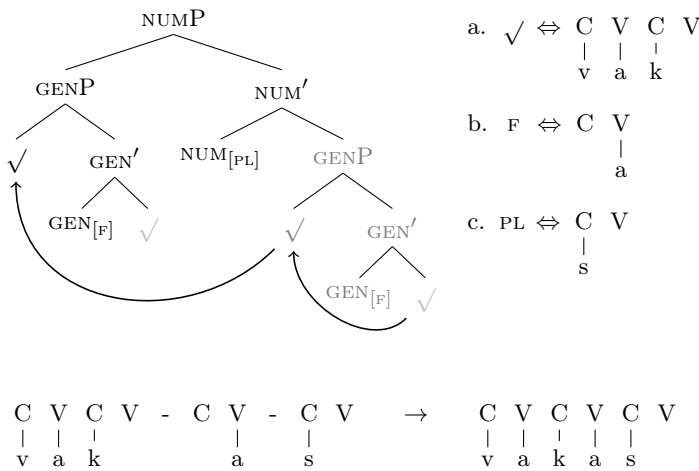
In Strict CV, this configuration is ill-formed, as empty V slots need to be properly governed, and they can only be properly governed by non-empty V slots (see Cavirani, 2022 and Youngberg, 2023 for a discussion of other conditions of proper government). As shown in (31)b, the non-empty V node of the F marker can properly govern the root-final V slot (as customary, proper government is

⁹ The discussion of possible formal explanations of such an absence is not central to the topic of this paper, so, in order not to unnecessarily overload the discussion, I do not address such issue.

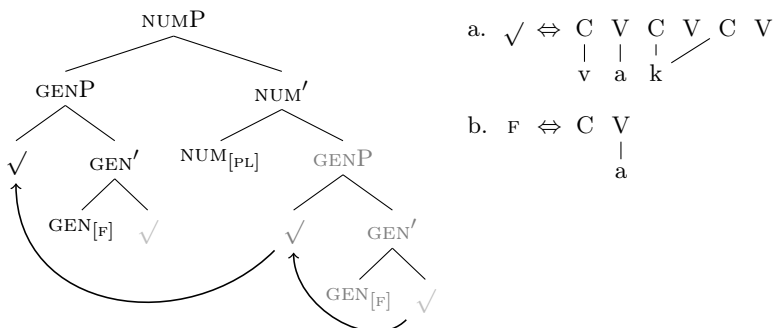
represented by a leftward arrow on top of the CV tier). As a consequence, the latter can stay silent. However, by virtue of being empty, this V slot cannot properly govern the preceding empty V slot, which thus needs to be pronounced. This triggers the epenthesis of [ə]. Were the PL marker associated to the root-final V slot, the latter would not be empty, and could properly govern the preceding V slot, generating *[ˈpegria], i.e. a structure with no epenthesis and an *ia* hiatus.

Crucially, the phonological derivation just sketched allows us to maintain one and the same derivation for Romance NUMPs, no matter how the GEN and NUM markers are linearly ordered. In this scenario, the variation between the varieties showing a different linear ordering is due to a difference in their phonological representations and their phonological grammar. This is shown below, where the syntactic derivation of the form for ‘cows’ in Spanish, Italian, and Colonnatese is given together with the phonological representation of the relevant markers.

(32) Spanish *vac-a-s* ‘cows’

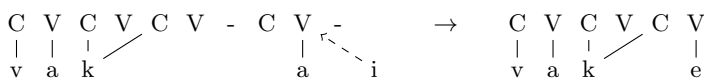
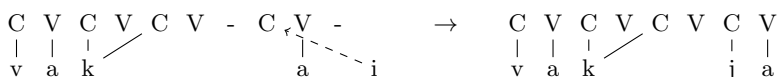
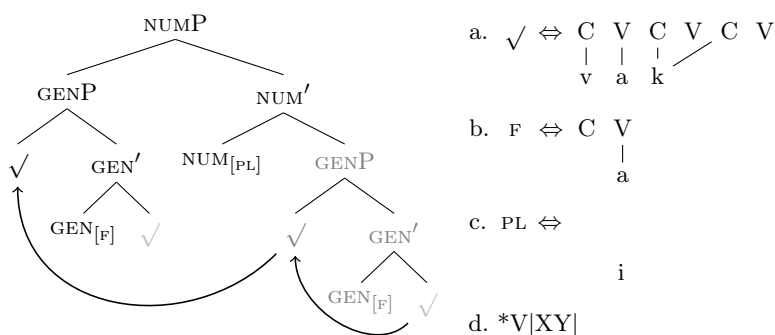


(33) Italian *vacch-e* ‘cows’



c. PL $\Leftrightarrow$

i

(34) Colonnatese *vacch-j-a* 'cows'

As shown in (32), (33), and (34), respectively, Spanish, Italian, and Colonnatese display the very same syntactic derivation, where the root cyclically moves to Spec,NUMP pied-piping GENP. They differ though with respect to the phonological representation of the GEN and NUM markers. In Spanish, both markers come with their CV slots, which allow them to surface independently, deriving ['vakas]. In Italian, the F marker has its own CV slots, whereas the PL marker is a floating [I] element, which associates to the V slot of of the F marker containing the [A] element, deriving [AI], i.e. [e], thus ['vak:e]. In Colonnatese, [AI] is ruled out by a constraint (absent in Italian phonological grammar) preventing unstressed nuclei to host complex element expressions (see *V|XY| in (34), where X and Y stand for any element; Cavirani, 2015). As a consequence, [I] associates to the empty C slot of the F marker, deriving ['vak:ja].¹⁰

¹⁰ An alternative phonological analysis can be developed where *V|XY| plays no role, and the difference between (32), (33), and (34) is defined in purely representational terms. Such

3.3 A short note on PC

Before concluding, I would like to briefly comment on PC in light of the phonological analysis just proposed. In such analysis, the F marker is phonologically represented as a CV structure, whose V slot contains /a/. In the Lunigianese varieties displaying PC, the PL marker – /-i/ – exploits the empty C slot of the F marker, surfacing as [-j]. In this scenario, depending on one's syntactic assumptions on the nature of concord, the absence of the PL marker on some constituent could be conceived of as due to some syntactic principle that, when a constituent occurs in specific structural configurations, either prevents the merger of the PL head, or prevents the copying of the PL feature/value on such constituent. In either case, the PL head/feature would be missing from the final syntactic representation. Namely, assuming the representations proposed in the previous section, the element [I] lexicalising PL on those specific constituents would never enter the morphosyntactic picture. Its phonological host – the C slot of the F marker – would be available, but unused.

Another way to conceive of PC would be relating it to the presence *vs* absence of the host of the floating [I], namely whether the CV unit that hosts /a/ in the V slot is present or not. This could be obtained by decomposing what we have identified as the F marker into two separate morphophonological objects, i.e. a CV structure and a floating [A]. In this case, the syntactic principle responsible for PC, rather than determining whether and where the PL marker enters the derivation, would determine the presence/absence of the CV structure hosting the floating exponents of F and PL. This is shown in the representations below, where the F (35) and M (36) Colonnatese forms for 'wolf' showing full concord, i.e. showing the extra CV structure, are compared with the F (37) and M (38) forms showing PC, i.e. the forms that do not feature the extra CV structure. These examples show the relevant forms in a sequence that reflects their syntactic derivation (in a framework that assumes that lexicalisation applies each time a head is merged, such as nanosyntax).

an analysis builds on the refinement of Strict CV representational component discussed in Cavarani & Vanden Wyngaerd (2026), which introduces root nodes. This allows for the distinction between floating elements and floating segments, i.e. root nodes that contain elements, but are not associated to any CV slot. Under this approach, the PL marker of Italian could be represented as a floating [I], as in (33), while the one of Colonnatese would be represented as a floating segment. Under the assumption that floating segments associate to empty CV slots, and floating elements associate to root nodes, the PL marker of Italian could only associate to the root node of the segment hosted in the V slot of F, while that of Colonnatese could only associate to its empty C slot.

(35) Full concord, F

- a. $\begin{array}{ccccccc} C & V & C & V & - & C & V \\ | & | & | & & & & \\ l & u & p & & & & \end{array} \quad \begin{array}{c} \swarrow \\ \text{a} \end{array}$
- b. $\begin{array}{ccccccc} C & V & C & V & - & C & V \\ | & | & | & & & & \\ l & u & p & & & & \end{array} \quad \begin{array}{c} \swarrow \\ \text{a} \end{array} \quad \begin{array}{c} \swarrow \\ \text{i} \end{array}$
- c. $\begin{array}{ccccccc} C & V & C & V & C & V \\ | & | & | & & | & | \\ l & u & p & & j & a \end{array}$

(37) Partial concord, F

- a. $\begin{array}{ccccccc} C & V & C & V & - \\ | & | & | & & \\ l & u & p & & \end{array} \quad \begin{array}{c} \swarrow \\ \text{a} \end{array}$
- b. $\begin{array}{ccccccc} C & V & C & V & - \\ | & | & | & | & \\ l & u & p & a & \end{array} \quad \begin{array}{c} \swarrow \\ \text{i} \end{array}$
- c. $\begin{array}{ccccccc} C & V & C & V \\ | & | & | & | \\ l & u & p & a & i \end{array}$

(36) Full concord, M

- a. $\begin{array}{ccccccc} C & V & C & V & - & C & V \\ | & | & | & & & & \\ l & u & p & & & & \end{array}$
- b. $\begin{array}{ccccccc} C & V & C & V & - & C & V \\ | & | & | & & & & \\ l & u & p & & & & \end{array} \quad \begin{array}{c} \swarrow \\ \text{i} \end{array}$
- c. $\begin{array}{ccccccc} C & V & C & V \\ | & | & | & | \\ l & u & p & i \end{array}$

(38) Partial concord, M

- a. $\begin{array}{ccccccc} C & V & C & V \\ | & | & | & \\ l & u & p & \end{array}$
- b. $\begin{array}{ccccccc} C & V & C & V & - \\ | & | & | & & \\ l & u & p & & \end{array} \quad \begin{array}{c} \swarrow \\ \text{i} \end{array}$
- c. $\begin{array}{ccccccc} C & V & C & V \\ | & | & | & | \\ l & u & p & i \end{array}$

As shown, the derivations given above neatly account for the concord patterns displayed by F nominal expressions. In full concord systems (35), both the F and the PL markers are pronounced, as they are both provided a landing slot by the extra CV structure. In PC systems (37), only the F marker is pronounced. This is because the F is the first one that is merged, and can thus associate to the final empty V slot made available by the root itself. The impossibility of the PL marker to surface is due to i) the absence of the extra CV structure providing an available landing site, and ii) the $*V[XY]$ constraint, which prevents the PL marker from associating with the root-final V slot hosting the F marker. In M forms, assuming that the M marker is null (or is not there; see Baggio, 2025), the PL marker can always associate, either to the V slot of the extra CV (36) or to the root final V slot (38). Note that, in such an analysis, the PL marker is always present in the morphosyntactic and phonological representation. It just fails to surface, but it still plays a role in the syntax. One consequence of this is that, despite not being pronounced, it can control subject-verb agreement, as in (17), repeated in (39), where the verb shows a PL marker despite no PL marker surfacing on the noun or on the demonstrative or the subject clitic.

- (39) kl-a pegr-a l ɛŋ l-a vɔstr-a
 that-F sheep-F SCL be.3PL the-F your-F
 ‘those sheep are yours’

One question that still begs for an answer is what the morphosyntactic correlate of the extra CV structure might be. I cannot provide any exhaustive answer, but I would like to suggest that its presence could be related to the internal phasal structure of the nominal expression, along the lines of Szigetvári (1999); Pagliano (2003); Scheer (2004); Seigneur-Froli (2006); Balogné-Bérce (2005); Kula & Marten (2009); Sanoudaki (2010); Cyrino & Espinal (2020); Newell (2025). Note that in such approach, Lunigianese PC does not necessarily support the Concord Hierarchy Generalization discussed in 1, as the absence of the absence of the PL marker in F forms is not due to the trimming of the top of the hierarchy in (1), but rather to the absence of the lower empty CV marker. Interestingly, though, the marker missing from the surface phonological structure is still the one lexicalising the top node of (1). I leave the exploration of this hypothesis for future research.

4 Conclusion

This paper proposes an analysis of an unexpected linear ordering of the GEN and NUM markers in a set of non-standard Romance varieties spoken in Lunigiana, Italy. This linear ordering is unexpected because it apparently violates the Mirror Principle (Baker, 1985), which predicts that the GEN marker is closer to the root than the NUM marker. This is in fact the linear ordering that we find in virtually all the Romance varieties that lexicalises GEN and NUM as two different markers, and arguably also in varieties which mark GEN and NUM oppositions by means of a change of the quality of the only marker occurring after the root (d’Hulst, 2006; Passino, 2009; Lampitelli, 2010, 2014; Lampitelli & Ulfsbjorninn, 2023). In a set of Lunigianese varieties, though, we find a different ordering, i.e. $\sqrt{\text{NUM-GEN}}$. In principle, this unexpected linear ordering could be given a morphosyntax-based explanations, which rely on the possibility of having multiple NUM projections (Manzini & Savoia, 2017), a relatively marked (within the Romance linguistic domain) syntactic derivation (Taraldsen, 2010; Muriungi, 2014; Myler, 2017), arbitrary morphological templates (Bloomfield, 1962; Zwicky, 1985; Anderson, 1986; Simpson & Withgott, 1986; Speas, 1990; Stump, 1991; Inkelas, 1993; Hyman & Inkelas, 1999; Good, 2003), postsyntactic operations such as Local Dislocation (Embick & Noyer, 2001; Arregi & Nevins, 2012), weighted bigram constraints encoding morphotactic restrictions (Ryan, 2010), specific rankings of morphophonological alignment and phonological

constraints (Kim, 2010; Zukoff, 2023), or phonological representations (Paster, 2009; Kim, 2015; McCann, 2025).

I provide an analysis of the Lunigianese linear ordering that is purely phonological, and show that the observed $\sqrt{\text{-NUM-GEN}}$ ordering, in fact, does not require a reordering of the corresponding syntactic heads, and can be neatly derived by refining the phonological representation of the relevant markers. This refinement is obtained by employing the formal tools provided by Strict CV (Lowenstamm, 1996; Scheer, 2004) and Element Theory (Bacley, 2011). This allows for developing an analysis in which the PL marker is a floating $[\text{I}]$ element that associates to the empty C slot of the CV structure hosting the F marker – an $[\text{A}]$ element, i.e. $[\text{a}]$ – in its V slot, surfacing as $[\text{j}]$. The impossibility of the $[\text{I}]$ element to join $[\text{A}]$ in the latter's V slot, deriving $[\text{AI}]$, i.e. $[\text{e}]$ (as happens in Italian), is due to a general phonological property of these varieties, which do not tolerate mid vowels in unstressed nuclei. These varieties do not tolerate unstressed falling diphthongs either, which prevents the possibility to generate $\sqrt{\text{-GEN-NUM}}$, i.e. $\sqrt{\text{-a-j}}$ (provided that one could epenthesise a CV structure to host the floating $[\text{I}]$, which I exclude on operation-economy grounds). On the other hand, complex onsets are very frequent, which makes of $\sqrt{\text{-j-a}}$ a perfectly acceptable string.

This analysis resonates with that proposed by Yu (2003, 2007) and Paster (2009), where affixes may subcategorize for other phonological objects, such as, in our case, C and V slots, and aligns with Paster (2009)'s claim that phonologically conditioned affix ordering does not exist, as “word-building proceeds from the ‘inside-out’ [thus] an affix cannot have any effect [...] on the location of another affix that was attached at an earlier stage in the derivation” (Paster, 2009, 24). Thus, the observed Lunigianese linear ordering should in fact be considered a case of segment metathesis that is triggered by the floating nature of the PL marker and the availability of a C slot in the F marker, rather than a case of affix metathesis (similarly to the Doyayo and Witsuwit'en cases discussed in Paster, 2009). In Paster (2009)'s words, it is “the association of the segmental feature to the CV tier, but not the affixation itself, [that] is handled by the phonology”. Crucially, “the location of the morpheme [...] in the morpho[syntactic] tree structure is unaffected by phonology”. The observed linear ordering can therefore be considered just an *apparent* case of phonologically-determined affix reordering.

Finally, my analysis i) provides support for a fully modular grammar architecture, where phonology directly interprets the structure generated by morphosyntax, as it does not require any postsyntactic movement operation nor any interleaving of morphosyntactic and phonological computation into one and the same grammar component (as required by analyses that combine morphophonological alignment and phonological constraints), and ii) is consistent with the hypothesis that the main locus of grammar variation is the Lexicon, namely, in this case, in the phono-

logical representation of the markers, as the difference in GEN and NUM marking between Italian, Spanish, and Lunigiana varieties can be derived if one assumes their markers have different phonological representations.

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